

Haptic Performance using Voltage-Mode Motor Control

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Abstract—Haptic devices driven by DC motors are usually controlled in current mode due to the direct relationship between current and torque. This work analyzes the performance of voltage-mode controllers whose main drawback is that the torque of the actuator depends on its electrical dynamics. However, the electrical dynamics of the motor add the viscosity generated by the back-electromotive force. Since the motor damping seen from the handle of the interface is increased by the square of the transmission ratio, the physical damping of the mechanism can be very high, maintaining low inertia. As a result, very high performance in terms of critical stiffness can be achieved, even using cost-effective electronics. There is a trade-off between the achievable virtual stiffness during the haptic interaction and the backdrivability in free movement, if the damping is set too high. To investigate the benefits of this motor control strategy, CEIT's haptic gearshift is used as a testbed. The experimental results confirm that a very high critical stiffness can be achieved using this strategy.

Index Terms—Control system analysis, Force control, Haptic interfaces, Mechanical variables control.

I. INTRODUCTION

HAPTIC systems map the movements of a user onto a virtual environment, and, if the user collides with virtual objects, they also provide force feedback [1]. This contact force should be high enough to avoid penetrating the virtual scene, and the resulting control loop is called “impedance control” [2]. Impedance causality means that the user can penetrate the virtual objects and the force feedback is calculated as a function of the penetration, typically by means of a viscoelastic force law (a spring-damper system).

The actuators placed at the joints of the interface produce the forces to be fed back to the user. The joint torques can be calculated using the Jacobian matrix of the mechanical interface. Electric DC motors are typically used for this purpose. Since these motors exert a torque proportional to the supplied current, the amplifiers are usually configured in *current mode*. These drivers have to sense the electrical current and they require a very fast internal current control loop.

A possible alternative is the use of dedicated amplifiers in *voltage mode*, which are simpler and potentially cheaper. The

main drawback of this solution is that the torque of the actuator depends on its electrical dynamics. For example, the current drawn by the motor decreases as electrical resistance changes due to variation in winding temperature. Another effect of the electrical dynamics of the motor is the viscosity added by the back-EMF [3]. This electrical damping has been used to increase the range of stable haptic impedances [4], [5].

The aim of this work is to investigate the performance of DC motors controlled in voltage mode. Since the motor damping translated to the handle of the interface is increased by the square of the transmission ratio [3], the physical damping of the mechanism can be very high, maintaining low inertia. As a result, very high performance in terms of critical stiffness can be achieved, even using cost-effective electronics. There is a trade-off between the virtual stiffness achievable during the haptic interaction and the backdrivability in free movement if the damping is set too high.

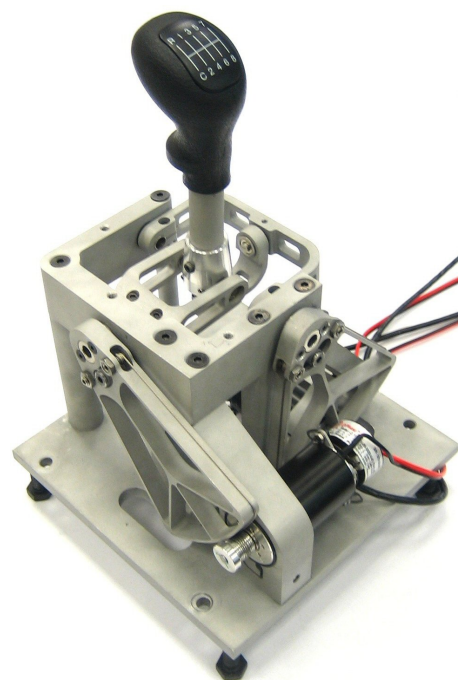


Fig. 1. Haptic gearshift mechanism

To analyze the benefits of the proposed motor control strategy, the haptic gearshift developed at CEIT [6] is used as an experimental testbed (Fig. 1). The main advantage of using this mechanism is that it has two decoupled degrees of freedom [6], and thus, it is possible to derive a very simple

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electromechanical model of each joint. Using this physical description of the joint, analyzing the influence of the motor control in voltage mode in the dynamics of the system and the performance of the impedance haptic loop is straightforward.

This paper is organized as follows. Section 2 describes the haptic loop in voltage mode and the physical parameters of the device. Section 3 shows the experimental critical stiffness of the system and a model of the system for predicting those critical values. The advantages of the voltage control mode with respect the current control mode and the influence of the temperature are discussed in Section 4. Finally, some conclusions are reported in Section 5.

II. SYSTEM DESCRIPTION

To describe the impedance haptic loop in voltage mode, a physical model of each joint of the CEIT haptic gearshift is first derived. This model includes the electrical description of the DC motor and some mechanical parameters of the joint. Fig. 2 shows the scheme of the joint. It is a motor-reducer-lever set where $\phi(t)$ is the rotation of the motor and $\theta(t)$ is the angle rotated by the lever. The transmission between the motor and the lever is depicted with two pulleys.

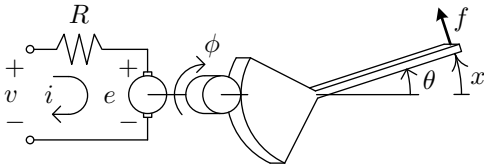


Fig. 2. Physical scheme of the joint

The input variable is the voltage signal $v(t)$, and the rest of the electrical signals are the current intensity $i(t)$ and the back-EMF voltage $e(t)$. The electrical resistance is R and the torque constant of the motor is κ . In the mechanical part of the system, $f(t)$ is the force exerted by the motor at the end of the lever. The output signal is the displacement $x(t)$, which results from multiplying angle $\theta(t)$ by the length of the lever d . The block diagram that relates all the signals in the Laplace domain is shown in Fig. 3.

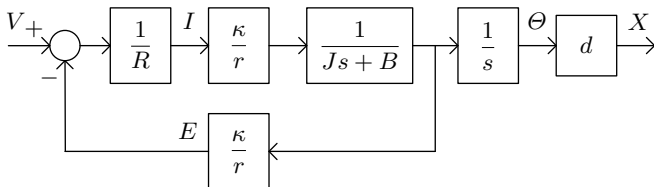


Fig. 3. Block diagram of the joint

Parameter J is the mechanical inertia and B the mechanical viscous damping. Both parameters are given in the rotational axis of the lever, and r is the reduction ratio of the transmission. Note that the effect of the inductance of the motor is neglected because the electrical time constant is much smaller

than the mechanical time constant. The transfer function that relates the input voltage $V(s)$ with the displacement $X(s)$ is:

$$X(s) = \frac{1}{\frac{J}{d^2}s^2 + \left(\frac{B}{d^2} + \frac{\kappa^2}{d^2 r^2 R}\right)s} \frac{\kappa V(s)}{rdR} \quad (1)$$

The equivalent mass of the system referred to the end of the lever along displacement x is $m = J/d^2$, while the equivalent mechanical damping is $b = B/d^2$. The back-EMF produces an additional viscous effect, called here b_e or electrical damping:

$$b_e = \frac{\kappa^2}{d^2 r^2 R} \quad (2)$$

Since the electrical resistance R changes due to Joule heating, the electrical damping b_e also changes with the temperature of the motor's winding. The force $f(t)$ exerted by the motor at the end of the lever is proportional to the input voltage $v(t)$, although R does change due to Joule heating:

$$f(t) = \frac{\kappa}{rdR} v(t) \quad (3)$$

Using previous definitions, an equivalent transfer function for the system is:

$$X(s) = \frac{1}{ms^2 + (b + b_e)s} F(s) \quad (4)$$

These expressions summarize two phenomena known in the literature, but not evident from just looking at Fig. 2, namely: i) the voltage supplied to the DC motor can produce a proportional force at the handle of the haptic device, and ii) the electrical dynamics of the motor adds a physical viscosity. Several characteristics of the motor and the mechanical transmission ratio affect the enumerated effects.

A. Impedance haptic loop using voltage

The relation between the force and the voltage (3) is used to implement a haptic impedance in voltage mode. Fig. 4 shows the haptic loop that computes an elastic force with stiffness K , while the position of the virtual wall is X_w . The diagram also shows that the loop manages discrete-time sampled signals, where T is the sampling period.

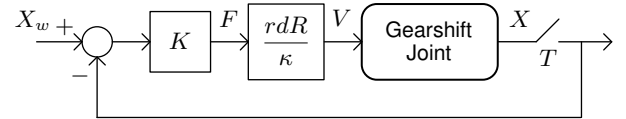


Fig. 4. Impedance haptic loop in voltage mode

A common goal in haptics is to implement virtual walls that are as stiff as possible in order to simulate perfectly rigid and immovable objects. The absolute upper bound of stiffness K is the critical limit of the system, that is, the stiffness coefficient that makes the system unstable. In the last decades, the most commonly used way to increase the range of stable stiffness coefficients is to implement virtual damping in the system [7]–[10]. However, hybrid electromechanical damping [11], [12] or only electrical damping [4], [5] is also used to increase the range of stable haptic impedances.

B. Joint physical parameters

It is difficult to know a priori which of the two damping coefficients— b or b_e —is dominant in system dynamics (4). To ensure high backdrivability in free movement, desktop devices are designed to have small mass m and mechanical damping b . Using the CAD model of the device, it is possible to estimate the inertia. However, it is difficult to determine the mechanical damping. In contrast, the electrical damping b_e is quite easy to estimate, because it depends on several parameters of the DC motor and the reduction ratio of the transmission (2).

Table I shows the specifications of several haptic interfaces [13]. In general, the mechanical damping coefficients are very small. In the case of the haptic gearshift, the first joint (which allows lateral movement from left to right and vice versa) will be used from here onwards for the discussion and the experiments. The length of the lever is $d = 0.15$ m, and the reduction ratio is $r = 1/16$. The actuators is a Maxon RE40 DC motor with torque constant $\kappa = 60.3$ mN·m/A. The electrical resistance of the circuit R is 1.6Ω . Thus, according to (2), the electrical damping b_e for the joint is 25.85 N·s/m.

TABLE I
MAIN CHARACTERISTICS OF SEVERAL HAPTIC DEVICES [13]

Device	Δx (μm)	m (kg)	b (N·s/m)	T (ms)	K_{CR} (N/m)
Impulse Engine	31.4	0.032	0.02	0.2	800
PHANTOM 1.0	29.1	0.072	0.005	1	1015
Freedom 6	20	0.25	0.01	1	2400
MIT Toolhandle	20.1	0.119	0.001	1	3125
Delta	30	0.25	0.01	0.33	14500
Omega	10	0.22	0.01	0.33	14500

This value is only an estimation because of the electrical resistance changes due to Joule heating. For example, the resistance R varies from 1.6Ω at 25°C , to 1.94Ω at 80°C . For the motor's winding at this last temperature, the electrical damping of the joint drops from 25.85 N·s/m to 21.29 N·s/m. Nevertheless, the electrical damping b_e is several orders of magnitude larger than the mechanical damping coefficients reported in Table I. Thus, it is reasonable to suppose that the mechanical damping is much smaller than the electrical damping ($b \ll b_e$).

Regarding the inertia of the joint, the contribution of the rotor is 134 g·cm², which is 152.46 g at the lever's handle. Using the CAD model of the haptic gearshift, the inertia of the device along the first joint is approximately $m \approx 0.3$ kg, which is comparable to some values reported in Table I. Thus, half of the overall inertia is due to the actuator's rotor.

The position resolution at the extreme of the gearshift lever is $\Delta x = 28.8 \mu\text{m}$, which is again comparable to the haptic interfaces listed in Table I.

III. EXPERIMENTAL RESULTS

The impedance algorithm in voltage mode is experimentally implemented and tested with the haptic gearshift (Fig. 1). The electronics consists of an Arduino Mega 2560 microcontroller board with a Pololu Dual VNH5019 motor shield [14]. This

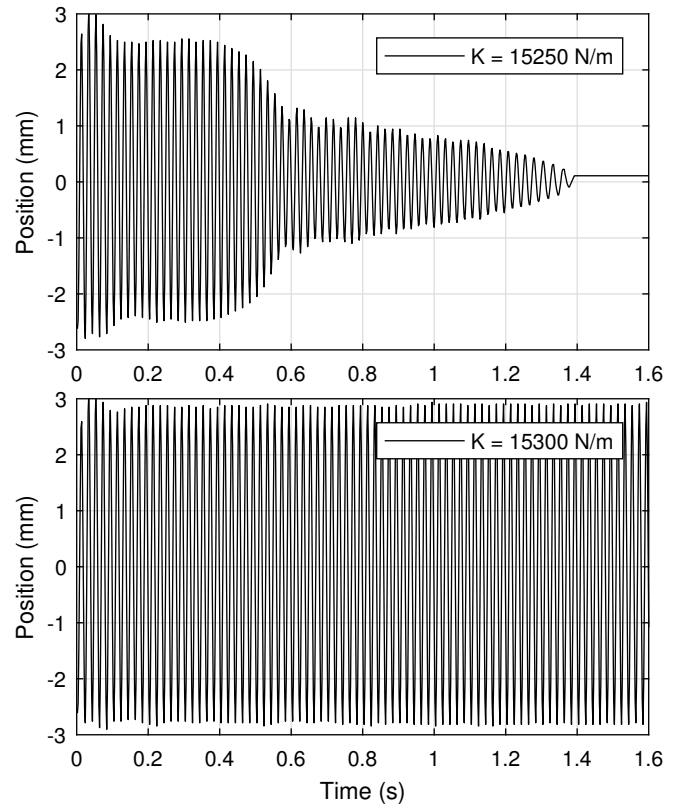


Fig. 5. Stable system response with stiffness $K = 15250$ N/m (top) and the critical response with $K = 15300$ N/m (bottom)

amplifier incorporates two full-bridge motor drivers, which are connected to a 12 V power supply. The default sampling period T is set to 2 ms. The controller cannot operate with sampling periods shorter than 1.5 ms because it also performs the encoder counts.

A bilateral virtual wall in the first joint of the gearshift is implemented, as illustrated in Fig. 4, where $x_w = 0$. To avoid the presence of the human dynamics in the control loop, which can introduce additional damping, mass, etc., the experiments are designed so that the system moves without user intervention. To achieve this, the position of the device is initialized with a value x_0 , which is distinct from the position of the bilateral wall x_w . To ensure that all the experiments have the same initial force of 40 N, x_0 is defined as follows:

$$x_0 = x_w - \frac{40}{K}. \quad (5)$$

To obtain the critical stiffness coefficient K_{CR} , the stiffness is increased until sustained oscillations around the reference position of the wall are observed. Using the default sampling period of 2 ms, Fig. 5 shows the last stable response of the gearshift by setting the stiffness to 15250 N/m, and the critical response for $K = 15300$ N/m.

The steady state in Fig. 5-top is not exactly zero, nor is it equal to the position resolution. This final position is due to the static friction, which is approximately 1.9 N in this joint.

The critical stiffness and the critical oscillation frequency change with the sampling rate and the overall time delay of the control loop [10]. Table II shows the values obtained

experimentally for several combinations of sampling period T and time delay t_d artificially added in the loop by the software.

TABLE II
CRITICAL RESPONSES OF THE HAPTIC GEARSHIFT

Sampling T (ms)	Delay t_d (ms)	Critical frequency ω_{CR} (rad/s)	Critical stiffness K_{CR} (N/m)
1.5	0	349.03	18400
2	0	314.16	15300
2.5	0	291.54	13700
3	0	261.82	11550
2	2	207.35	7620
2	4	174.99	5950
2	6	156.77	5650
2	8	71.94	3400
2	10	65.97	2230

Although longer delays could be introduced by software, the oscillation amplitude could reach the mechanical limits of the mechanism.

Since relatively slow sampling rates imply relatively small critical stiffness coefficients [7], [10], the experimental results of $K_{CR} = 15300$ N/m with $T = 2$ ms and $K_{CR} = 18400$ N/m with $T = 1.5$ ms are striking. In fact, these two values are larger than the critical limits presented in Table II.

A. Prediction of the stability limits

Under certain assumptions, it should be possible to predict the critical stiffness of the haptic interaction. The stability criterion reported in [10] states that:

$$\text{Critical stiffness} = \frac{\sum \text{Damping}}{\sum \text{Delay}} \quad (6)$$

Using the physical parameters of the gearshift, and assuming that the electrical damping is much larger than the mechanical damping ($b_e \gg b$), the stability criterion can be expressed as follows:

$$K_{CR} = \frac{b_e}{\frac{T}{2} + t_d + \tau}, \quad (7)$$

where t_d is reserved for the time delay artificially added in the loop by the software and τ is the inherent delay of the electronic hardware [15].

Taking the critical values in Table II and the estimation of the electrical damping, $b_e \approx 25.85$ N·s/m, the only unknown parameter in (7) is τ , which in principle should be a constant smaller than the sampling period. Substituting in (7) the critical stiffness coefficients K_{CR} with $t_d = 0$, the obtained inherent delay τ is around $650 \mu\text{s}$. Although this is a plausible result, the theoretical criterion does not predict all the critical coefficients reported in Table II well.

The experimental oscillation frequencies ω_{CR} can help us know why the prediction of the critical values is not possible. As reported in [10], the oscillation frequency of the system depends on the effective stiffness, damping and mass:

$$\omega_{CR} = \sqrt{\frac{K_{CR}}{m} - \frac{b_e^2}{4m^2}}. \quad (8)$$

By substituting in (8) the critical stiffness coefficients K_{CR} with $t_d = 0$ and parameter b_e , the resulting experimental mass for the system is around 150 g. This is approximately the inertia of the rotor (half of the complete mechanism). This means that the device does not behave like a rigid body and that the cable transmission is decoupling the mechanical system in two masses (Fig. 6).

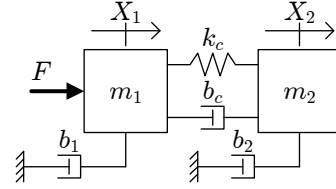


Fig. 6. Mechanical model with two connected masses

The mechanical model of haptic devices with two connected masses is introduced in [16]. As shown in Fig. 6, besides the two masses, three damping coefficients and the stiffness coefficient modeling the cable can be considered. Fitting these six mechanical parameters and the unknown time delay τ due to the electronics, it is possible to obtain a good model for the system. For example, Fig. 7 shows the experimental data and the frequency response of the model with two connected masses, using the parameters of Table III.

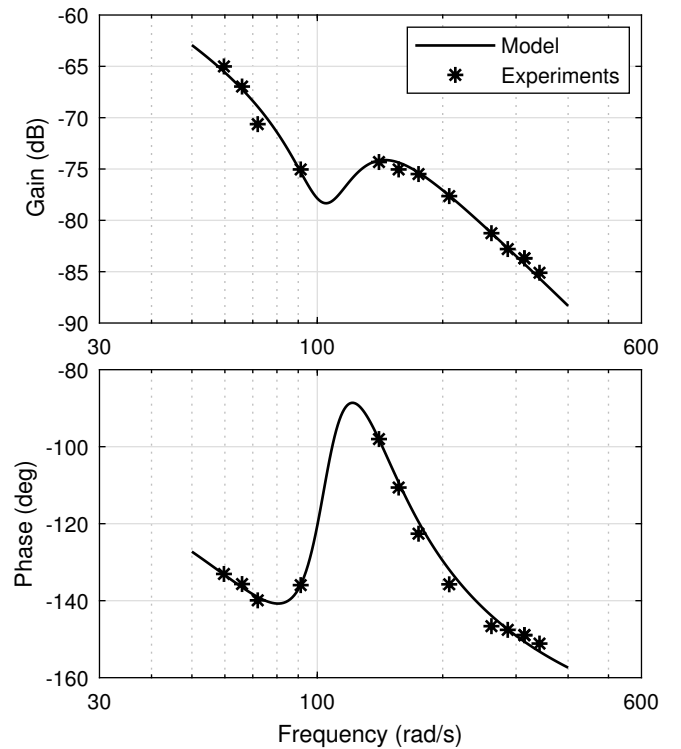


Fig. 7. Frequency response of the model with two connected masses and experimental results

The identification method for fitting the parameters reported in Table III is similar to the optimization work presented in [15]. The translation of the experimental data of critical limits into the frequency response of the system is given by the

TABLE III
FITTED PARAMETERS OF THE GEARSHIFT

Parameter	Variable	Fitted value
Device mass 1	m_1	0.161 kg
Device damping 1	b_1	21.66 N·s/m
Cable stiffness	k_c	1416 N/m
Cable damping	b_c	3.28 N·s/m
Device mass 2	m_2	0.128 kg
Device damping 2	b_2	0.11 N·s/m
Time delay	τ	719 μ s

following equations:

$$\text{Gain (dB)} = -20 \log K_{CR} \quad (9)$$

$$\text{Phase (rad)} = -\pi + \left(\frac{T}{2} + t_d + \tau \right) \omega_{CR} \quad (10)$$

The mechanical model with two connected masses fits what we know about certain physical parameters of the system: m_1 fits well the mass of the rotor, b_1 with the electrical damping, and m_2 with the mass estimated by the CAD file. This model also explains the experimental critical oscillations better than the linear criterion (7).

B. Response with the user moving the device

The above results are completed with a set of experimental responses of the system, but now with the user grasping and pushing the lever. Fig. 8 shows two stable responses, where the stiffness coefficient is set to 16000 N/m and 18125 N/m.

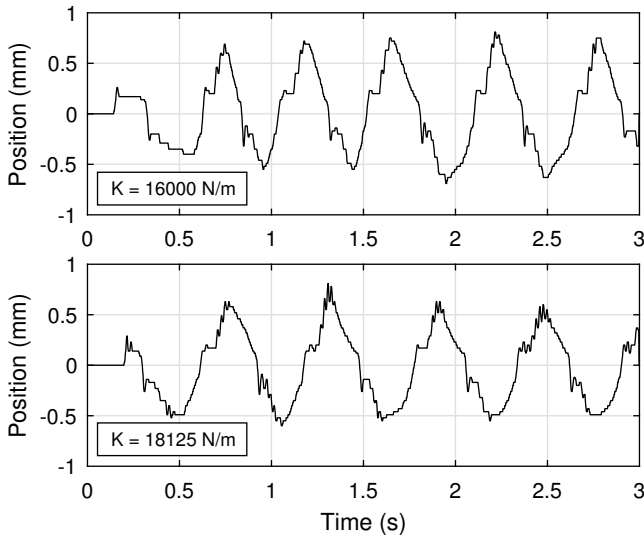


Fig. 8. Stable system response with stiffness $K = 16000$ N/m (top) and $K = 18125$ N/m (bottom)

Although the experiments with the human operator are done with the default sampling period of $T = 2$ ms, stable responses are registered with stiffness coefficients much larger than 15300 N/m (Table II). This is consistent with [7], [17], which states that the presence of human arm dynamics—physical stiffness, damping and mass—contributes to stability and increases the range stable haptic impedances.

Fig. 9 shows the response of the system with a stiffness coefficient equal to 21750 N/m. Although the response seems to be stable during the first second, it becomes unstable.

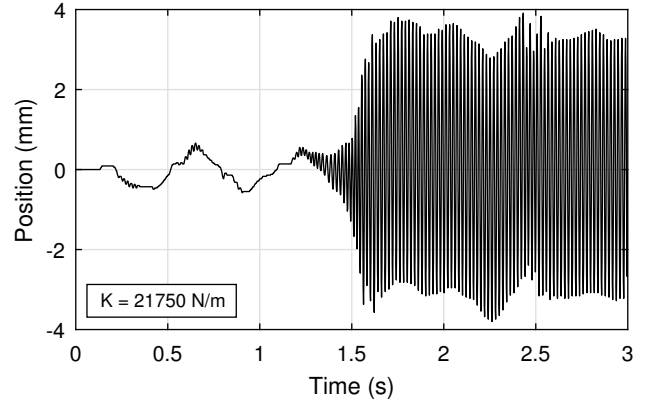


Fig. 9. Unstable system response with stiffness $K = 21750$ N/m

By definition, unstable systems produce unbounded output signals for any bounded input. Thus, instability can be detected by the presence of undamped and increasing oscillations (time interval $1.2 < t < 1.6$ s in Fig. 9). This situation may reach sustained oscillations with constant and large amplitude if the actuators reach saturation ($t > 1.6$ s in Fig. 9).

IV. DISCUSSION

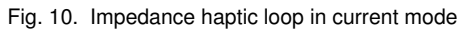
The human tactile sense has some limitations, which can be used to select the appropriate stiffness coefficients for the haptic interaction. Several authors have estimated the stiffness threshold beyond which a virtual wall is perceived as perfectly rigid. This threshold could be called the *maximum perceivable stiffness*. Massie and Salisbury set this threshold at 2000 N/m [18], while Tan et al. establishes this limit at around 24200 N/m [19]. Since it is possible to implement stable stiffness coefficients around this threshold, it is clear that the control strategy using DC motors in voltage mode achieves excellent performance even using cost-effective electronics.

A. Current-mode vs. voltage-mode control mode

To analyze the main differences between the two controlling options, Fig. 10 shows the block diagram of an elastic haptic impedance K , but now using current-mode control. The current loop usually includes a PI controller, but in this example only a proportional gain K_p is used. The main fact that illustrates this block diagram is that the back-EMF is seen as a disturbance within the current loop that has to be rejected.

The transfer function that relates the displacement $X(s)$ and the force of the user $F_h(s)$ in the Laplace domain is:

$$X(s) = \frac{1}{\frac{J}{d^2} s^2 + \left[\frac{B}{d^2} + \frac{\kappa^2}{d^2 r^2 (R + K_p)} \right] s + \frac{KK_p}{R + K_p}} F_h(s) \quad (11)$$



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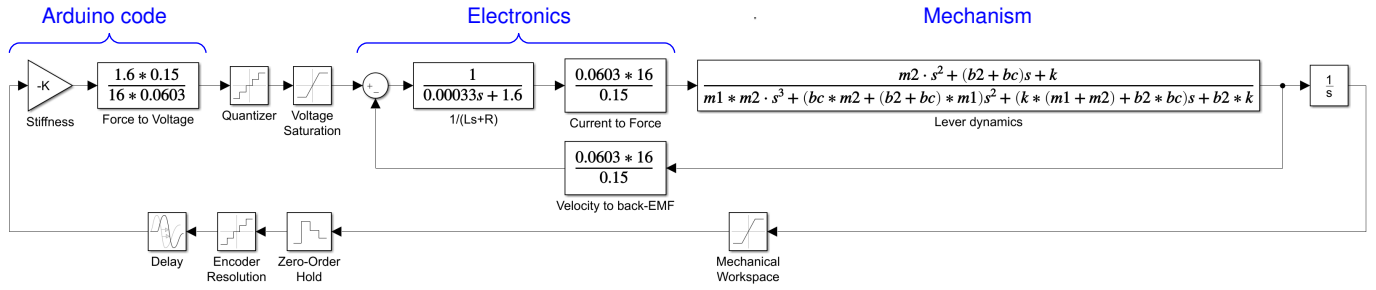


Fig. 11. Simulink model of the system

C. Influence of the sampling rate

The effect of the sampling period on stability and critical stiffness can also be shown with discrete-time simulations using the Simulink model of the system. Table V presents the range of sampling periods simulated, the frequency of the critical oscillations and the critical stiffness coefficients.

TABLE V
CRITICAL STIFFNESS VS. SAMPLING PERIOD

Sampling period T (ms)	Critical frequency ω_{CR} (rad/s)	Critical stiffness K_{CR} (N/m)
0.5 ms	383.68 rad/s	24170 N/m
1 ms	348.40 rad/s	20188 N/m
1.5 ms	321.38 rad/s	17415 N/m
2 ms	299.60 rad/s	15361 N/m
2.5 ms	281.55 rad/s	13785 N/m
3 ms	266.40 rad/s	12550 N/m

Sampling periods shorter than 1.5 ms, not feasible with our electronic hardware, get very high critical stiffness coefficients. However, they are also located at high frequencies, outside the experimental Bode diagram (Fig. 7), and therefore, unmodeled dynamics of the mechanism could change these two critical gains. The non-linear phenomena and the electrical inductance included in the Simulink model (Fig. 11) but not fitted experimentally, cause a small difference between the simulated critical oscillations and the experimental results (Table II).

V. CONCLUSION

When controlling mechanical joints by means of DC motors in voltage mode, relatively high physical damping is caused by the back-electromotive force. In haptic applications, this viscous effect can be used to make the system more stable. This paper analyses this hypothetical benefit and shows experimentally that the critical stiffness can be very high even when using cost effective electronics and a desktop haptic interface.

In the case of our haptic interface, the stability limit—without additional delay in the loop—is placed over the vibration mode of the mechanical transmission that decouples the system in two masses, and it is therefore not possible to obtain a linear relation between the critical stiffness and the viscous damping. Nevertheless, the theoretical model of the interface using two connected masses predicts the stability limits well.

The value of the electric damping can be estimated quite well simply by using the known parameters of the electric

motor and the complete multiplication ratio of the joint. Thus, useful information about the behavior of the system may be predicted during the design phase of the mechanical interface, and thus engineers can select the motors and the transmission ratio that guarantees a certain amount of damping.

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